

ASG TEMPLATE VALIDATION REPORT

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C. | Standard R1 | master 01_Master_Template/ASG_Master_Template.dxf (sha256 1eac00933ddd0895...) | tools: ezdxf 1.4.4, LibreOffice 24.2.7.2 420(Build:2) | CAD software found: none | 10.10.2026

Summary: PASS 150 | PARTIAL 1 | UNVERIFIED 19 | FAIL 2. AutoCAD was NOT available: no test below ran inside AutoCAD. PASS = tested by re-opening the saved file; PARTIAL = tested with a non-AutoCAD renderer; UNVERIFIED = needs AutoCAD (ASG-QA and README step 6 cover each one); FAIL = test failed or deliverable missing.

FAIL items: Deliverables: ASG_Master_Template.dwt generated natively; Deliverables: ASG_Master_Template.dwg generated natively

Test procedure

1. Inventory CAD software in the environment (AutoCAD, accoreconsole, ODA, LibreDWG, QCAD, BricsCAD).
2. Build every file from asg_standard.py (build_all.py).
3. Re-open the master with ezdxf recover + audit, strict parse, and LibreOffice Draw as an independent reader.
4. Compare every header variable (incl. current dimension variables), layer, linetype, text / dim / mleader / multiline style and annotation scale with the definitions.
5. Check page settings, borders, title-block position / scale / attributes / locks, viewport layer / lock / scale / position, ByLayer properties of every sheet object.
6. Check model space empty, no xrefs / underlays / proxies, only title-block blocks, fonts, linetypes, hatch pattern names (acadiso.pat), ISO 128 / 3098 / 216 series.
7. Idempotency: re-run scale injection and layout creation; rebuild and compare; check the generated LISP.
8. Re-open each single-size template and the CTB (binary header + all 255 entries).
9. Separate temporary QA drawing: dimensions of known geometry, viewports 1:20 / 1:5, no-plot objects, lineweight matrix, text heights. Check measured values and displayed precision; render and measure lineweights.
10. Pending in AutoCAD: ASG-SETUP / ASG-QA (0 FAIL), AutoCAD PDF plot of the QA drawing, annotative scaling test, SAVEAS DWG / DWT.

Results

Area	Test	Status	Evidence
Deliverables	ASG_Master_Template.dwt generated natively	FAIL	Not generated: no DWG writer in this environment (acad, accoreconsole, ODAFileConverter, TeighaFileConverter, dwg2dxf, dxf2dwg, qcad, librecad, bricscad all absent). Not faked by renaming. Created in AutoCAD by README step 5.
Deliverables	ASG_Master_Template.dwg generated natively	FAIL	Not generated: no DWG writer in this environment (acad, accoreconsole, ODAFileConverter, TeighaFileConverter, dwg2dxf, dxf2dwg, qcad, librecad, bricscad all absent). Not faked by renaming. Created in AutoCAD by README step 5.
File	Master DXF re-opened from disk (recover + audit)	PASS	0 errors, 0 fixes, AC1032 = AutoCAD 2018 DXF
File	Master DXF strict parse	PASS	ezdxf.readfile without exception
File	Master DXF opened by an independent second reader	PASS	LibreOffice 24.2.7.2 420(Build:2)
File	Opens in AutoCAD; ASG-QA reports 0 FAIL	UNVERIFIED	AutoCAD not available - README step 3-4
Units & variables	\$INSUNITS = 4	PASS	read back 4
Units & variables	\$MEASUREMENT = 1	PASS	read back 1
Units & variables	\$LUNITS = 2	PASS	read back 2
Units & variables	\$LUPREC = 0	PASS	read back 0
Units & variables	\$AUNITS = 0	PASS	read back 0

Area	Test	Status	Evidence
Units & variables	\$AUPREC = 2	PASS	read back 2
Units & variables	\$ANGBASE = 0.0	PASS	read back 0.0
Units & variables	\$ANGDIR = 0	PASS	read back 0
Units & variables	\$LIMMIN = (0.0, 0.0)	PASS	read back (0.0, 0.0)
Units & variables	\$LIMMAX = (200000.0, 150000.0)	PASS	read back (200000.0, 150000.0)
Units & variables	\$LIMCHECK = 0	PASS	read back 0
Units & variables	\$LTSCALE = 1.0	PASS	read back 1.0
Units & variables	\$PSLTSCALE = 1	PASS	read back 1
Units & variables	\$CELTSCALE = 1.0	PASS	read back 1.0
Units & variables	\$LWDISPLAY = 1	PASS	read back 1
Units & variables	\$CELWEIGHT = -1	PASS	read back -1
Units & variables	\$CECOLOR = 256	PASS	read back 256
Units & variables	\$DIMASSOC = 2	PASS	read back 2
Units & variables	\$PLINEGEN = 1	PASS	read back 1
Units & variables	\$FILLMODE = 1	PASS	read back 1
Units & variables	\$MIRRTEXT = 0	PASS	read back 0
Units & variables	\$TEXTSIZE = 2.5	PASS	read back 2.5
Units & variables	\$PSTYLEMODE = 1	PASS	read back 1
Units & variables	\$WORLDVIEW = 1	PASS	read back 1
Units & variables	\$UCSORG = (0.0, 0.0, 0.0)	PASS	read back (0.0, 0.0, 0.0)
Units & variables	\$UCSXDIR = (1.0, 0.0, 0.0)	PASS	read back (1.0, 0.0, 0.0)
Units & variables	\$UCSYDIR = (0.0, 1.0, 0.0)	PASS	read back (0.0, 1.0, 0.0)
Units & variables	\$VISRETAIN = 1	PASS	read back 1
Units & variables	\$PROXYGRAPHICS = 1	PASS	read back 1
Units & variables	\$CLAYER = ASG-OUTLINE-PRIMARY	PASS	read back ASG-OUTLINE-PRIMARY
Units & variables	\$TEXTSTYLE = ASG-TEXT-ANNO	PASS	read back ASG-TEXT-ANNO
Units & variables	\$DIMSTYLE = ASG-DIM-ANNO	PASS	read back ASG-DIM-ANNO
Units & variables	\$CMLSTYLE = ASG-MLINE-2	PASS	read back ASG-MLINE-2
Units & variables	Header dimension variables = ASG-DIM-ANNO (no '<style overrides>')	PASS	43 variables compared
Units & variables	Drawing variables set by ASG-SETUP (CANNOSCALE, MSLTSCALE, HPLAYER ...)	UNVERIFIED	Not writable by DXF libraries; ASG-QA checks them in AutoCAD
Units & variables	Profile variables (OSMODE, grips, autosave ...)	UNVERIFIED	Registry, not drawing-resident; applied only by opt-in ASG-PROFILE
Linetypes	HIDDEN loaded	PASS	
Linetypes	HIDDEN2 loaded	PASS	
Linetypes	CENTER loaded	PASS	
Linetypes	CENTER2 loaded	PASS	
Linetypes	DASHED loaded	PASS	
Linetypes	DASHED2 loaded	PASS	
Linetypes	DASHDOT loaded	PASS	
Linetypes	DASHDOT2 loaded	PASS	
Linetypes	PHANTOM loaded	PASS	
Linetypes	PHANTOM2 loaded	PASS	
Linetypes	DOT loaded	PASS	
Linetypes	Continuous loaded	PASS	

Area	Test	Status	Evidence
Layers	34 ASG layers present	PASS	found 34
Layers	No other layers except system 0 / Defpoints	PASS	['0', 'Defpoints']
Layers	Layer names and codes unique	PASS	
Layers	Every layer: ACI, linetype (loaded), linewidth, plot flag, description	PASS	34/34 correct
Layers	No-plot layers exactly: ASG-CONSTRUCTION, ASG-NOPLOT, ASG-VIEWPORT	PASS	
Layers	Current layer is a plottable geometry layer	PASS	ASG-OUTLINE-PRIMARY
Lineweights	Plotted lineweights from ISO 128 series (0.09 hatch only)	PASS	[0.09, 0.13, 0.18, 0.25, 0.35, 0.5]
Lineweights	Every plotted layer in a linewidth-matrix class	PASS	
Text styles	ASG-TEXT-ANNO: arial.ttf, height 0, width 1, annotative=True	PASS	
Text styles	ASG-TEXT-NOTE: arial.ttf, height 0, width 1, annotative=True	PASS	
Text styles	ASG-TEXT-SMALL: arial.ttf, height 0, width 1, annotative=True	PASS	
Text styles	ASG-TEXT-TITLE: arialbd.ttf, height 0, width 1, annotative=False	PASS	
Text styles	ASG-TEXT-HEADING: arialbd.ttf, height 0, width 1, annotative=False	PASS	
Text styles	ASG-TEXT-SHEET: arial.ttf, height 0, width 1, annotative=False	PASS	
Text styles	Recommended plotted heights in ISO 3098 series	PASS	
Text styles	Annotative text behaviour at viewport scales	UNVERIFIED	Annotative flag written and re-read; scaling needs AutoCAD (README step 6b)
Dimension styles	ASG-DIM-ANNO	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-FIXED	PASS	all 44 variables match; annotative=False
Dimension styles	ASG-DIM-DETAIL	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-ANGULAR	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-RADIUS	PASS	all 44 variables match; annotative=True
Dimension styles	ASG-DIM-DIAMETER	PASS	all 44 variables match; annotative=True
Dimension styles	'Standard' dim style made harmless (= ASG-DIM-FIXED values)	PASS	
Multileader styles	ASG-ML-ANNO	PASS	ASG-TEXT-NOTE 2.5 mm, arrow closed filled 2.5
Multileader styles	ASG-ML-NOTE	PASS	ASG-TEXT-SHEET 2.5 mm, arrow closed filled 2.5
Multileader styles	ASG-ML-DETAIL	PASS	ASG-TEXT-SMALL 1.8 mm, arrow_DOT 1.5
Multileader styles	ASG-ML-REFERENCE	PASS	ASG-TEXT-NOTE 2.5 mm, arrow_OPEN30 2.5
Multileader styles	Background mask	UNVERIFIED	Mask is an MText property, not a style property - per-note procedure documented
Multiline styles	ASG-MLINE-2	PASS	
Multiline styles	ASG-MLINE-3	PASS	
Annotation scales	Scale list = 1:1, 1:2, 1:5, 1:10, 1:20, 1:25, 1:50, 1:100	PASS	read back ['1:1', '1:2', '1:5', '1:10', '1:20', '1:25', '1:50', '1:100'], paper:drawing units verified
Annotation scales	No duplicate scales	PASS	
Annotation scales	Scale list accepted by AutoCAD	UNVERIFIED	ASG-QA re-checks it in AutoCAD
Table styles	7 ASG-TABLE-* styles	UNVERIFIED	No DXF library can write TABLESTYLE; created by ASG-SETUP. In DXF now: none
Page setups	Named page setups ASG-<layout>	UNVERIFIED	Named PLOTSETTINGS not writable here; ASG-SETUP copies them from the layouts
Layouts	Layouts: A4-LANDSCAPE, A4-PORTRAIT, A3-LANDSCAPE, A3-PORTRAIT, A1-LANDSCAPE	PASS	['Model', 'A4-LANDSCAPE', 'A4-PORTRAIT', 'A3-LANDSCAPE', 'A3-PORTRAIT', 'A1-LANDSCAPE']
Sheets	A4-LANDSCAPE: ISO 216 size 297 x 210	PASS	
Layouts	A4-LANDSCAPE: DWG To PDF.pc3, media, Layout area, 1:1 mm, offset 0,0, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A4_(297.00_x_210.00_MM)
Layouts	A4-LANDSCAPE: paper media name recognised by DWG To PDF.pc3	UNVERIFIED	ASG-QA checks it against the device's media list in AutoCAD
Title blocks	A4-LANDSCAPE: ASG-TB-A4 at scale 1 (not stretched) on border corner, 15 attributes	PASS	
Sheets	A4-LANDSCAPE: border, 20 mm filing margin / 10 mm others	PASS	

Area	Test	Status	Evidence
Viewports	A4-LANDSCAPE: 1 viewport, display locked, ASG-VIEWPORT (no plot), 1:10	PASS	
Viewports	A4-LANDSCAPE: viewport inside border and clear of title block	PASS	
Sheets	A4-PORTRAIT: ISO 216 size 210 x 297	PASS	
Layouts	A4-PORTRAIT: DWG To PDF.pc3, media, Layout area, 1:1 mm, offset 0,0, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A4_(210.00_x_297.00_MM)
Layouts	A4-PORTRAIT: paper media name recognised by DWG To PDF.pc3	UNVERIFIED	ASG-QA checks it against the device's media list in AutoCAD
Title blocks	A4-PORTRAIT: ASG-TB-A4 at scale 1 (not stretched) on border corner, 15 attributes	PASS	
Sheets	A4-PORTRAIT: border, 20 mm filing margin / 10 mm others	PASS	
Viewports	A4-PORTRAIT: 1 viewport, display locked, ASG-VIEWPORT (no plot), 1:10	PASS	
Viewports	A4-PORTRAIT: viewport inside border and clear of title block	PASS	
Sheets	A3-LANDSCAPE: ISO 216 size 420 x 297	PASS	
Layouts	A3-LANDSCAPE: DWG To PDF.pc3, media, Layout area, 1:1 mm, offset 0,0, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A3_(420.00_x_297.00_MM)
Layouts	A3-LANDSCAPE: paper media name recognised by DWG To PDF.pc3	UNVERIFIED	ASG-QA checks it against the device's media list in AutoCAD
Title blocks	A3-LANDSCAPE: ASG-TB-A3 at scale 1 (not stretched) on border corner, 15 attributes	PASS	
Sheets	A3-LANDSCAPE: border, 20 mm filing margin / 10 mm others	PASS	
Viewports	A3-LANDSCAPE: 1 viewport, display locked, ASG-VIEWPORT (no plot), 1:20	PASS	
Viewports	A3-LANDSCAPE: viewport inside border and clear of title block	PASS	
Sheets	A3-PORTRAIT: ISO 216 size 297 x 420	PASS	
Layouts	A3-PORTRAIT: DWG To PDF.pc3, media, Layout area, 1:1 mm, offset 0,0, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A3_(297.00_x_420.00_MM)
Layouts	A3-PORTRAIT: paper media name recognised by DWG To PDF.pc3	UNVERIFIED	ASG-QA checks it against the device's media list in AutoCAD
Title blocks	A3-PORTRAIT: ASG-TB-A3 at scale 1 (not stretched) on border corner, 15 attributes	PASS	
Sheets	A3-PORTRAIT: border, 20 mm filing margin / 10 mm others	PASS	
Viewports	A3-PORTRAIT: 1 viewport, display locked, ASG-VIEWPORT (no plot), 1:20	PASS	
Viewports	A3-PORTRAIT: viewport inside border and clear of title block	PASS	
Sheets	A1-LANDSCAPE: ISO 216 size 841 x 594	PASS	
Layouts	A1-LANDSCAPE: DWG To PDF.pc3, media, Layout area, 1:1 mm, offset 0,0, ASG_Monochrome.ctb, plot lineweights	PASS	ISO_full_bleed_A1_(841.00_x_594.00_MM)
Layouts	A1-LANDSCAPE: paper media name recognised by DWG To PDF.pc3	UNVERIFIED	ASG-QA checks it against the device's media list in AutoCAD
Title blocks	A1-LANDSCAPE: ASG-TB-A1 at scale 1 (not stretched) on border corner, 15 attributes	PASS	
Sheets	A1-LANDSCAPE: border, 20 mm filing margin / 10 mm others	PASS	
Viewports	A1-LANDSCAPE: 1 viewport, display locked, ASG-VIEWPORT (no plot), 1:50	PASS	
Viewports	A1-LANDSCAPE: viewport inside border and clear of title block	PASS	
Title blocks	ASG-TB-A4: 15 attributes, 180 x 68.2 mm, own design	PASS	
Title blocks	ASG-TB-A4: block units mm, not explodable, uniform scale, attributes position-locked	PASS	
Title blocks	ASG-TB-A4: text heights >= 1.8 mm on paper	PASS	[1.8, 2.5, 3.5]
Title blocks	ASG-TB-A3: 15 attributes, 180 x 81.9 mm, own design	PASS	
Title blocks	ASG-TB-A3: block units mm, not explodable, uniform scale, attributes position-locked	PASS	
Title blocks	ASG-TB-A3: text heights >= 1.8 mm on paper	PASS	[1.8, 2.5, 3.5, 5.0]

Area	Test	Status	Evidence
Title blocks	ASG-TB-A1: 15 attributes, 250 x 109.2 mm, own design	PASS	
Title blocks	ASG-TB-A1: block units mm, not explodable, uniform scale, attributes position-locked	PASS	
Title blocks	ASG-TB-A1: text heights >= 1.8 mm on paper	PASS	[2.5, 3.5, 5.0, 7.0]
Title blocks	A4 / A3 / A1 title blocks are separate designs (row heights differ)	PASS	ASG-TB-A4: 68.2 mm, ASG-TB-A3: 81.9 mm, ASG-TB-A1: 109.2 mm
Properties	All layout / title-block objects ByLayer (colour, linetype, lineweight)	PASS	[]
Model space	Model space empty	PASS	0 entities
References	No external references	PASS	
References	No image / PDF underlay dependencies	PASS	
Interoperability	No proxy / custom objects	PASS	
Scope	Only title-block blocks (no product blocks / libraries)	PASS	['ASG-TB-A1', 'ASG-TB-A3', 'ASG-TB-A4']
Interoperability	Fonts referenced: Arial TrueType only	PASS	['arial.ttf', 'arialbd.ttf']
Interoperability	Every layer linetype embedded in the file (no .lin dependency)	PASS	
Hatch standard	All hatch pattern names exist in AutoCAD acadiso.pat	PASS	[]
Automation	Re-running scale injection changes nothing	PASS	
Automation	Rebuild produces identical layer / style / layout sets	PASS	
Automation	Re-running layout creation creates no duplicate layouts	PASS	
Automation	ASG_Setup.lsp generated: balanced, covers every layer / style / layout / variable	PASS	73 names checked
Automation	ASG_Setup.lsp runs in AutoCAD (ASG-SETUP / ASG-QA)	UNVERIFIED	Written without AutoCAD; get-or-create logic, read-only QA
Drawing templates	ASG_A4-LANDSCAPE.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A4-PORTRAIT.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A3-LANDSCAPE.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A3-PORTRAIT.dxf: opens, single layout, full standard, empty model	PASS	
Drawing templates	ASG_A1-LANDSCAPE.dxf: opens, single layout, full standard, empty model	PASS	
Plot style	ASG_Monochrome.ctb: 60-byte header (3 x uint32) + zlib body re-read	PASS	ezdxf writer defect on 64-bit Linux (8-byte fields) worked around and verified
Plot style	ACI 1-249,255 -> black; 250-254 -> object colour (grey); object lineweight & linetype	PASS	
Plot style	CTB opens in AutoCAD Plot Style Table Editor with 'Use object lineweight'	UNVERIFIED	Lineweight code per ezdxf/acad.ctb convention; ASG-QA checks the CTB is installed; README step 2 checks the editor; fallback monochrome.ctb
Lineweights	Lineweight matrix rendered at 2400 dpi: 6 classes distinct, ordered, proportional	PARTIAL	rendered/nominal: 0.50:1.02, 0.35:1.03, 0.25:0.96, 0.18:1.00, 0.13:0.92, 0.09:1.00 (1.00 = exact). Hierarchy proven; absolute plotted mm needs an AutoCAD PDF plot.
QA drawing	Temporary QA drawing is a separate file	PASS	QA_TEST_DRAWING_temporary.dxf
Dimensions	Linear dims measure true geometry 1200 / 2400	PASS	[1200.0, 2400.0]
Dimensions	Detail dim measures 50	PASS	
Dimensions	Angular dim measures 30 deg	PASS	30.0
Dimensions	Radius / diameter dims measure R40 / D80	PASS	R [40.0] D [80.0]
Dimensions	No overridden dimension text	PASS	
Dimensions	Displayed text: 1200 / 2400 (0 dp), 50.0 (detail), 30.00 deg, R40, D80	PASS	['1200', '2400', '50.0', '30.00°', 'R40', 'Ø80']
Viewports	QA viewports at 1:20 and 1:5	PASS	[5.0, 20.0]
Plotting	No-plot layers (ASG-CONSTRUCTION, ASG-NOPLOT) excluded from the plot	UNVERIFIED	Plot flags set and re-read (Layers: PASS). The preview renderer ignores plot flags, so the diagonal and 'NOPLOT' text are visible in QA_render_A3 - confirm absence in the AutoCAD PDF (README 6a)
Plotting	Viewport boundaries not plotted	UNVERIFIED	ASG-VIEWPORT is no-plot and 'plot viewport borders' off (PASS); renderer cannot show plot behaviour
Plotting	PDF via DWG To PDF.pc3 + CTB: margins, clipping, lineweights	UNVERIFIED	QA_render_A3.pdf is a preview, not an AutoCAD plot (README 6a)

Area	Test	Status	Evidence
Annotation scales	Annotative text / dims / mleaders across viewport scales	UNVERIFIED	No DXF library writes annotative scale contexts; test in AutoCAD (README 6b)
Deliverables	ASG_Layer_Register.xlsx written and re-opened (all layers listed)	PASS	
Deliverables	ASG_CAD_Standards.pdf written	PASS	

QA render - temporary QA drawing (A3, viewports 1:20 and 1:5, lineweight matrix, text heights). Preview only; not part of the master template. The preview renderer ignores layer plot flags, so no-plot objects are visible here.

TEXT 2.5 @1:20

2400

NO PLOT

1200

R40

Ø80

50.0

30.0

0.50 Extra heavy - ASG-SECTION-CUT

0.35 Heavy - ASG-OUTLINE-PRIMARY

0.25 Medium - ASG-ALUMINIUM-PROFILE

0.18 Light - ASG-OUTLINE-SECONDARY

0.13 Fine - ASG-GASKET

0.09 Hatch - ASG-HATCH

2.5 mm General dimensions

2.5 mm Technical notes

2.5 mm Material labels

1.8 mm Small detail notes

3.5 mm Detail titles

5.0 mm Drawing titles

7.0 mm Sheet headings

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.		
AL SIRAJ GROUP ALUMINIUM, GLASS & FACADE DIVISION UNITED ARAB EMIRATES		
PROJECT	LOCATION	
PROJECT NAME	LOCATION, UAE	
CLIENT	CONSULTANT	
CLIENT	CONSULTANT	
DRAWING TITLE		
DRAWING TITLE		
DRAWN BY	CHECKED BY	APPROVED BY
-	-	-
SCALE	DATE	DRAWING STATUS
AS SHOWN	DD.MM.YYYY	PRELIMINARY
DRAWING No.		SHEET No.
AS-YYYY-NNN-DIS-TYP-001		01 OF 01